Year 7

REVISION PACK

Unit 2

Name: _____

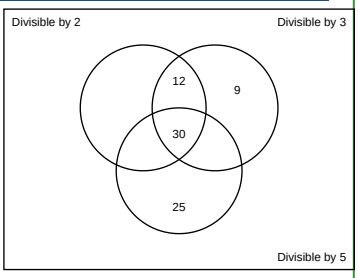
Class: _____

Unit 2 · Venn diagrams

Worked example: sorting numbers in a Venn diagram

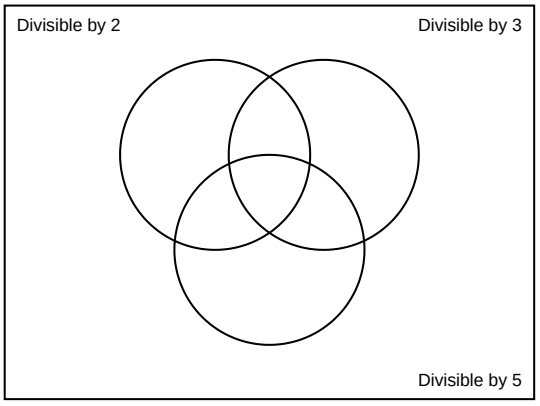
Put these numbers in the Venn diagram: 30 12 25 9

Step 1	Test 30: goes in all 3	$\div 2\checkmark \quad \div 3\checkmark \quad \div 5\checkmark$
Step 2	Test 12	$\div 2\checkmark \quad \div 3\checkmark \rightarrow$ 2 and 3 only
Step 3	Test 25	$\div 5\checkmark \rightarrow$ 5 only
Step 4	Test 9	$\div 3\checkmark \rightarrow$ 3 only
Step 5	Answer	write each number in its place



Practice: sorting numbers in a Venn diagram

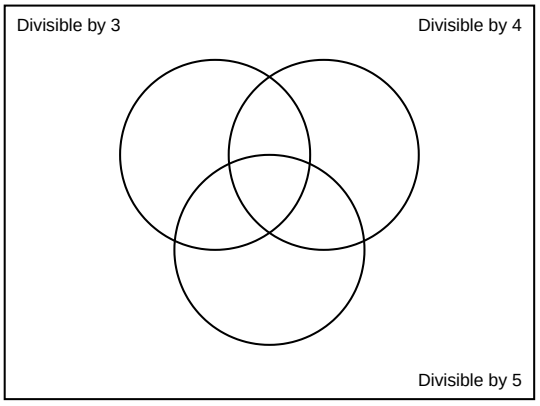
1.1



Put these numbers in the Venn diagram.

60 18 35 14

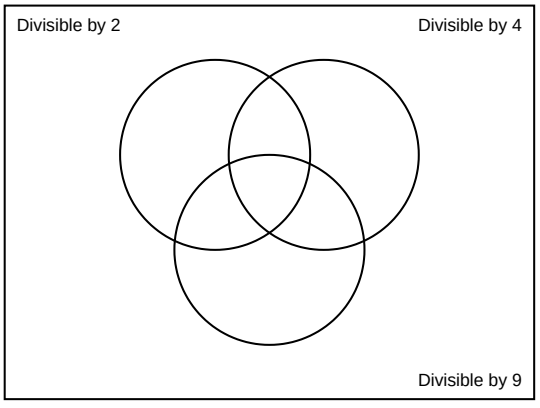
1.2



Put these numbers in the Venn diagram.

45 20 36 7

1.3

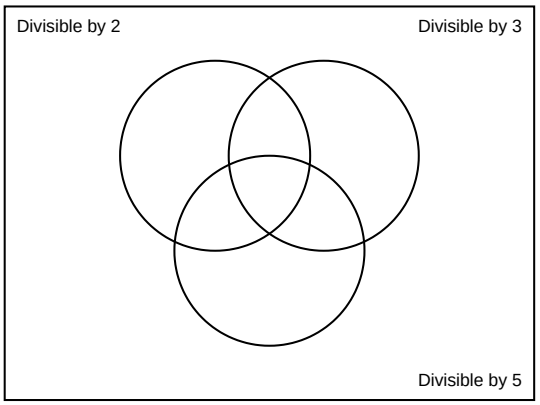


Put these numbers in the Venn diagram.

72 24 27 10

Practice: sorting numbers in a Venn diagram (continued)

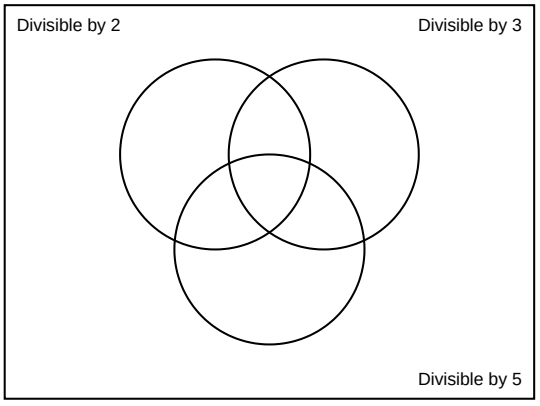
1.4



Put these numbers in the Venn diagram.

90 15 16 21

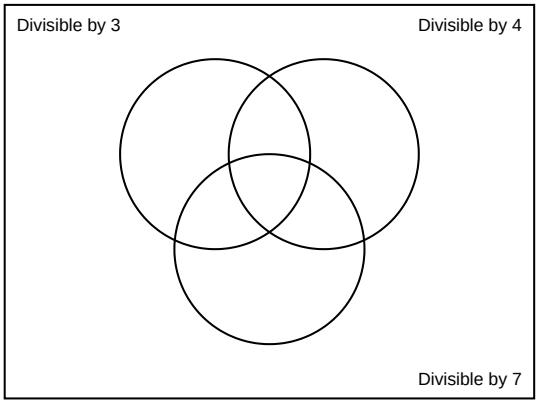
1.5



Put these numbers in the Venn diagram.

120 28 33 50

1.6



Put these numbers in the Venn diagram.

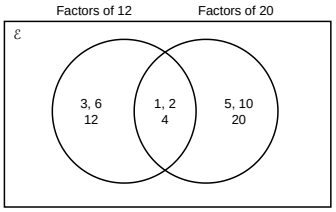
84 36 63 11

Worked example: factors in a Venn diagram, HCF and LCM

The numbers 12 and 20.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

(a)	Step 1	Factors of 12	$1, 2, 3, 4, 6, 12$
	Step 2	Factors of 20	$1, 2, 4, 5, 10, 20$
	Step 3	In both: the middle	$1, 2, 4$
	Step 4	Answer: the rest go outside the middle	
(b)	Step 1	Biggest number in the middle	4
	Step 2	Answer	4
(c)	Step 1	Multiples of 12	$12, 24, 36, 48, 60$
	Step 2	Multiples of 20	$20, 40, 60$
	Step 3	Answer	60



Practice: factors in a Venn diagram, HCF and LCM

2.1 The numbers 18 and 30.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

2.2 The numbers 12 and 18.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

2.3 The numbers 24 and 36.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

2.4 The numbers 20 and 28.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

2.5 The numbers 16 and 40.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

2.6 The numbers 15 and 35.

(a) Draw a Venn diagram to show their factors. (b) Write down the HCF. (c) Work out the LCM.

Worked example: calculating with negative numbers

Work out

(a) $-4-(-7)$ (b) 5×-6 (c) $-18\div-3$ (d) $2+(-9)$

(e) The temperature at night was -5°C. By noon it had risen by 9°C. Work out the temperature at noon.

(a)	**Step 1**	Minus a minus is plus	$-4-(-7)=-4+7=3$
(b)	**Step 1**	Signs different: answer is −	$5\times-6=-30$
(c)	**Step 1**	Signs the same: answer is +	$-18\div-3=6$
(d)	**Step 1**	Plus a minus is minus	$2+(-9)=2-9=-7$
(e)	**Step 1**	Rise means add	$-5+9=4$
	Step 2	Answer	$4\,^{\circ}\mathrm{C}$

Practice: calculating with negative numbers

3.1 Work out

(a) $-4-(-9)$ (b) 6×-7 (c) $-32\div-4$ (d) $7+(-15)$

(e) The temperature was -12°C. It rose by 15°C. Find the new temperature.

3.2 Work out

(a) $-7-(-3)$ (b) -8×5 (c) $54\div-6$ (d) $3+(-11)$

(e) The temperature was -9°C. It rose by 13°C. Find the new temperature.

3.3 Work out

(a) $5-(-6)$ (b) -9×-4 (c) $-45\div9$ (d) $-2+(-8)$

(e) The temperature was 2°C. It fell by 10°C. Find the new temperature.

3.4 Work out

(a) $-12-(-5)$ (b) 7×-7 (c) $-60\div-10$ (d) $8+(-20)$

(e) The temperature was -15°C. It rose by 9°C. Find the new temperature.

3.5 Work out

(a) $-2-(-10)$ (b) -3×-11 (c) $36\div-4$ (d) $-6+(-9)$

(e) The temperature was -18°C. It rose by 12°C. Find the new temperature.

3.6 Work out

(a) $-9-(-14)$ (b) 11×-2 (c) $-81\div9$ (d) $12+(-19)$

(e) The temperature was 6°C. It fell by 14°C. Find the new temperature.

Worked example: squares, cubes, powers of 10 and roots

Work out

(a) 300^2 (b) 0.2^3 (c) 10^5 (d) $\sqrt{144}$ (e) $\sqrt[3]{64}$

(a)	Step 1	Times itself	$3 \times 3 = 9$, add 4 zeros
	Step 2	Answer	$90\,000$
(b)	Step 1	Times itself 3 times	$2 \times 2 \times 2 = 8$
	Step 2	3 decimal places	0.008
(c)	Step 1	1 then 5 zeros	$100\,000$
(d)	Step 1	What times itself is 144?	$12 \times 12 = 144$
	Step 2	Answer	12
(e)	Step 1	What cubed is 64?	$4 \times 4 \times 4 = 64$
	Step 2	Answer	4

Practice: squares, cubes, powers of 10 and roots

4.1 Work out

(a) 500^2 (b) 0.4^3 (c) 10^4 (d) $\sqrt{121}$ (e) $\sqrt[3]{27}$

4.2 Work out

(a) 800^2 (b) 0.3^3 (c) 10^6 (d) $\sqrt{169}$ (e) $\sqrt[3]{125}$

4.3 Work out

(a) 400^2 (b) 0.1^3 (c) 10^3 (d) $\sqrt{81}$ (e) $\sqrt[3]{1000}$

4.4 Work out

(a) 900^2 (b) 0.6^3 (c) 10^5 (d) $\sqrt{196}$ (e) $\sqrt[3]{8}$

4.5 Work out

(a) 200^2 (b) 0.5^3 (c) 10^7 (d) $\sqrt{225}$ (e) $\sqrt[3]{343}$

4.6 Work out

(a) 600^2 (b) 0.7^3 (c) 10^2 (d) $\sqrt{100}$ (e) $\sqrt[3]{216}$

Unit 2 · Order of operations

Worked example: order of operations with powers and roots

Work out

(a) 3×4^2 (b) $5^2 - 2 \times \sqrt{49}$ (c) $2 \times (\sqrt{20 + 5} + 3)$ (d) $(2 + 3 \times 2)^2$

(a)	Step 1	Power first	$4^2 = 16$
	Step 2	Answer	$3 \times 16 = 48$
(b)	Step 1	Power and root first	$25 - 2 \times 7$
	Step 2	Then multiply	$25 - 14$
	Step 3	Answer	11
(c)	Step 1	Inside the root first	$\sqrt{25} = 5$
	Step 2	Bracket	$5 + 3 = 8$
	Step 3	Answer	$2 \times 8 = 16$
(d)	Step 1	Bracket: multiply first	$3 \times 2 = 6, \ \ 2 + 6 = 8$
	Step 2	Answer	$8^2 = 64$

Practice: order of operations with powers and roots

5.1 Work out

(a) 6×5^2 (b) $9^2 - 3 \times \sqrt{16}$ (c) $4 \times (\sqrt{1 + 8} + 5)$ (d) $(1 + 2 \times 3)^2$

5.2 Work out

(a) 2×9^2 (b) $8^2 - 4 \times \sqrt{25}$ (c) $3 \times (\sqrt{7 + 9} + 2)$ (d) $(4 + 1 \times 5)^2$

5.3 Work out

(a) 5×3^2 (b) $7^2 - 6 \times \sqrt{4}$ (c) $5 \times (\sqrt{12 + 4} + 6)$ (d) $(3 + 2 \times 4)^2$

5.4 Work out

(a) 4×6^2 (b) $10^2 - 9 \times \sqrt{81}$ (c) $2 \times (\sqrt{30 + 6} + 4)$ (d) $(2 + 3 \times 3)^2$

5.5 Work out

(a) 7×2^2 (b) $6^2 - 5 \times \sqrt{9}$ (c) $6 \times (\sqrt{50 + 14} + 1)$ (d) $(1 + 4 \times 2)^2$

5.6 Work out

(a) 3×7^2 (b) $11^2 - 8 \times \sqrt{100}$ (c) $10 \times (\sqrt{40 + 9} + 3)$ (d) $(5 + 1 \times 4)^2$

Worked example: estimating by rounding to 1 significant figure

Work out an estimate of $\dfrac{19 \times 41}{3.8}$

Step 1	Round each number to 1 s.f.	$19 \to 20,\ \ 41 \to 40,\ \ 3.8 \to 4$
Step 2	Multiply the top	$20 \times 40 = 800$
Step 3	Divide	$800 \div 4 = 200$
Step 4	Answer	200

Practice: estimating by rounding to 1 significant figure

6.1 Work out an estimate of

(a) $\dfrac{22 \times 29}{5.9}$ (b) $\dfrac{48 \times 19}{2.1}$ (c) $\dfrac{62 \times 29}{5.8}$

6.2 Work out an estimate of

(a) $\dfrac{39 \times 51}{2.2}$ (b) $\dfrac{18 \times 33}{3.1}$ (c) $\dfrac{79 \times 22}{8.3}$

6.3 Work out an estimate of

(a) $\dfrac{11 \times 58}{1.9}$ (b) $\dfrac{31 \times 42}{6.1}$ (c) $\dfrac{52 \times 68}{7.2}$

6.4 Work out an estimate of

(a) $\dfrac{27 \times 41}{3.9}$ (b) $\dfrac{69 \times 19}{1.8}$ (c) $\dfrac{38 \times 52}{4.1}$

6.5 Work out an estimate of

(a) $\dfrac{88 \times 12}{2.4}$ (b) $\dfrac{59 \times 41}{3.1}$ (c) $\dfrac{23 \times 47}{1.9}$

6.6 Work out an estimate of

(a) $\dfrac{42 \times 18}{4.1}$ (b) $\dfrac{71 \times 33}{7.4}$ (c) $\dfrac{49 \times 61}{5.2}$

Unit 2 · Factors and multiples problems

Worked example: sum of factors and LCM problems

(a) Show that the sum of the factors of 15 is the same as $5^2 - 1$.

(b) A lamp is every 60 m and a bench is every 45 m along a path. They are together at the start. How far along will they next be together?

(a)	**Step 1**	Factors of 15	$1, 3, 5, 15$
	Step 2	Add them	$1 + 3 + 5 + 15 = 24$
	Step 3	Work out the other side	$5^2 - 1 = 25 - 1 = 24$
	Step 4	Answer	$\text{both are } 24$
(b)	**Step 1**	Multiples of 60	$60, 120, 180$
	Step 2	Multiples of 45	$45, 90, 135, 180$
	Step 3	First in both: LCM	180
	Step 4	Answer	$180\ \text{m}$

Practice: sum of factors and LCM problems

7.1 (a) Show that the sum of the factors of 32 is the same as $4^3 - 1$.

(b) There is a distance marker every 100 m and a pole every 80 m. They are together at the start. How far along will they next be together?

7.2 (a) Show that the sum of the factors of 10 is the same as 3×6.

(b) There is a tree every 40 m and a lamp post every 30 m. They are together at the start. How far along will they next be together?

7.3 (a) Show that the sum of the factors of 21 is the same as 2^5.

(b) There is a bin every 50 m and a sign every 75 m. They are together at the start. How far along will they next be together?

7.4 (a) Show that the sum of the factors of 12 is the same as $5^2 + 3$.

(b) There is a bus stop every 120 m and a post box every 90 m. They are together at the start. How far along will they next be together?

7.5 (a) Show that the sum of the factors of 9 is the same as $3^2 + 4$.

(b) There is a red tile every 24 cm and a blue tile every 36 cm. They are together at the start. How far along will they next be together?

7.6 (a) Show that the sum of the factors of 16 is the same as $2^5 - 1$.

(b) There is a red bead every 18 cm and a gold bead every 30 cm. They are together at the start. How far along will they next be together?